

# X-ray binary populations in local galaxies

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**Resumen** / Las binarias de rayos X podrían haber contribuido significativamente a la inyección de energía en su entorno durante el Amanecer Cósmico, participando en la regulación de la tasa de formación estelar cósmica y la ionización del medio intergaláctico. Aún no hay una determinación precisa de su contribución debido principalmente al poco conocimiento de las propiedades de las poblaciones de binarias a alto *redshift*. Se han desarrollado modelos basados en la síntesis de poblaciones de binarias, pero un ingrediente clave se ha simplificado: la relación entre las propiedades de las binarias y su luminosidad se calcula mediante recetas simples. En este trabajo presentamos una prescripción original para las luminosidades, basada en la solución de las ecuaciones magnetohidrodinámicas de discos finos no estacionarios. Esta prescripción mejora el acuerdo de los resultados de los modelos de síntesis poblacional con las observaciones en rayos X de las galaxias en el Universo local.

**Abstract** / X-ray binaries may have contributed significantly to the injection of energy into the environment during the Cosmic Dawn, participating in the regulation of the cosmic star formation and the ionization state of the intergalactic medium. A precise determination of their contribution is still lacking mainly due to the poor knowledge about the properties of X-ray binary populations at high redshifts. Models based on population synthesis methods have been devised, but a crucial ingredient is usually overlooked: the link between the properties of the binaries and their X-ray luminosity is computed from simple recipes. In this work we present a novel prescription for X-ray luminosities, based on the solution of the magnetohydrodynamical equations of unsteady thin disks. This prescription improves the match of the results of population synthesis models with X-ray observations of galaxies in the Local Universe.

**Keywords** / X-rays: binaries — accretion, accretion disks — methods: numerical

## 1. Introduction

Some binary stellar systems eventually reach a stage in their evolution in which the primary is either a neutron star or a black hole, and the companion is a non-collapsed star. A strong stellar wind or the overflow of the Roche lobe of the latter, result in a disk through which matter loses angular momentum and is accreted by the former (Frank et al., 2002; Fabbiano, 2006, and references therein). Accretion extracts from the gravitational field up to half the rest energy of the infalling matter, heating the disk (cf. fusion releases only 0.7% of the hydrogen rest energy). The distinctive property of these systems is their large X-ray luminosity ( $10^{36-39}$  erg s<sup>-1</sup>) emitted by the inner, hot ( $10^{6-7}$  K) region of the disk, which explains their names: X-ray binaries (XRBs). They also eject part of the transferred matter through disk winds and relativistic jets (Mirabel et al., 1992), that may inflate bubbles in their neighbourhood (Feng & Soria, 2011) and produce cosmic rays (Heinz & Sunyaev, 2002; Escobar et al., 2021, 2022).

Among XRBs, special attention has to be paid to those in which the companion is a massive star

( $\gtrsim 8 M_{\odot}$ ), namely high-mass X-ray binaries (HMXBs). These systems accrete mainly through the strong stellar wind of the companion, and dominate young XRB populations (ages  $\lesssim 50$  Myr) hosted by star-forming galaxies. The end product of their evolution is a system of two compact objects, which may become a source of gravitational waves detectable by LIGO/Virgo/KAGRA. Although HMXBs are transient sources, their effective emission time (0.1 – 1 Myr, derived from the observed number of sources) implies that their total radiative energy release may reach up to  $\sim 10^{51}$  erg. A similar amount of kinetic energy may be released through massive outflows (Feng & Soria, 2011). This large energy feedback may play an important role in the modification of the environment.

Several works have found a trend showing an increase in the size and luminosity of HMXB populations with decreasing metallicity (Kaaret et al., 2011; Brorby et al., 2014; Douna et al., 2015; Ponnada et al., 2020), concluding that the effects of the aforementioned energy feedback would be more noticeable during the Cosmic Dawn. Indeed, the contribution of HMXBs to the regulation of the cosmic star formation, and to the heat-

ing and reionization of the intergalactic medium in that epoch (Mirabel et al., 2011; Fragos et al., 2013; Artale et al., 2015; Douna et al., 2018), has already been explored. However, its importance is still a matter of debate, due to the lack of complete knowledge about the four ingredients needed to construct a HMXB feedback model: 1) the star formation rate and metallicity distribution of galaxies; 2) the population of massive binaries in a galaxy; 3) the evolution of these binaries; 4) the energy release produced by accretion when active.

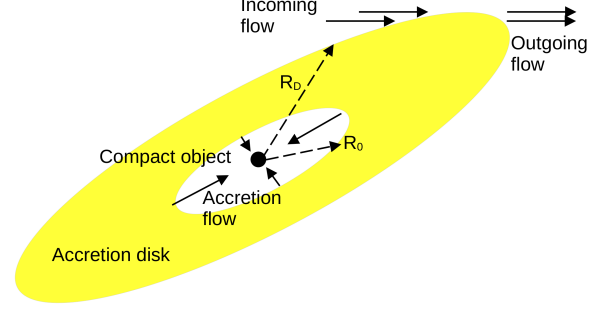
The first ingredient is given by both simulations of galaxy formation and evolution (e.g. Ceverino et al., 2017, 2018) and recent observations (Bouwens et al., 2015, 2016). However, the aforementioned metallicity trend suggests that the major role is played by low-mass, low-luminosity galaxies, whose properties are less constrained by both observations and simulations. The second one is provided by observations of local stellar and binary populations Chabrier (2003); Sana et al. (2012), whose validity for the conditions prevailing at Cosmic Dawn is uncertain. The third one is the most certain, as the theory of stellar and binary evolution is mature and has been contrasted with a large body of observations. However, specific issues like the outcome of the common envelope phase are still being debated.

The last ingredient is the most uncertain; previous works assume that the mass accretion rate is given by the Bondi & Hoyle (1944) formula, and that the X-ray luminosity is a fraction  $\eta \sim 0.1$  of the available accretion power. This overestimates both the size and luminosity of HMXB populations, which is solved by assuming that only a fraction  $f_{\text{em}}$  of the binaries are emitting at any given time (e.g. Douna et al., 2015). This semiempirical model neglects the fact that the infalling matter carries angular momentum, which must be transported out of the system for accretion to proceed. Therefore, part of the infalling matter must be expelled, and the rest should form an accretion disk to get rid of its angular momentum (Balbus & Hawley, 1998). The standard model (Shakura & Sunyaev, 1973) assumes a geometrically thin, optically thick disk accreting in a steady state. Although it describes correctly individual XRB luminosities, for typical parameters it predicts disk masses that imply formation time scales longer than the observed emission times, suggesting that the transient period should not be neglected. Indeed, the fraction  $f_{\text{em}}$  may be related to the formation and destruction time scales of the disk.

In this work, we focus on the last issue raised above by constructing a self-consistent model for the X-ray luminosity of an unsteady accretion disk. We describe our method in Sect. 2, and discuss our preliminary results in Sect. 3.

## 2. Unsteady disk model for HMXB feedback

Our goal is to develop a model for HMXB emission to be used in the computation of the energy feedback produced by the populations of these sources present in star-forming galaxies. In this first stage we concentrate in radiative feedback, testing our model against the X-ray luminosities of local galaxies compiled by Douna



**Fig. 1.** Scenario of our model. Part of the matter reaching the disk is expelled at the disk outer radius, carrying out the angular momentum transported outwards by the disk.

et al. (2015). The model includes only the last three ingredients described in Sect. 1, because we apply it to individual galaxies. The properties of these galaxies (star formation rate, metallicity) are taken from Douna et al. (2015). For the initial binary populations in each galaxy we adopt the initial mass function of Chabrier (2003), and the binary fraction and orbital parameter distributions provided by Sana et al. (2012). For each galaxy in the compilation of Douna et al. (2015), we create a simulated initial binary population by Monte Carlo sampling. Assuming that the star formation rate of these galaxies has not changed with time  $t$  for the last  $\Delta t = 50$  Myr (an interval larger than the maximum time required for a massive binary to transform into a double compact object system), we assign each binary a birth time in  $[-\Delta t, 0]$ , and compute its evolution up to the end of this interval with the code SEVN (Iorio et al., 2023). The end product of this simulation is a snapshot of the binary population of each galaxy at present.

From the simulated present binary population, we take those systems comprised by a black hole or neutron star and a non-collapsed star. For each system, using Eqns. 15–18 of Iorio et al. (2023), we compute the four main parameters describing wind accretion, namely the mass  $\dot{M}_{\text{acc}}$  and radius  $R_{\text{acc}}$  of the accretor, and the accretion rates of mass  $\dot{M}_{\text{acc}}^{\text{BH}}$  and angular momentum  $\dot{J}_{\text{acc}}^{\text{BH}}$  (here BH stands for Bondi-Hoyle).

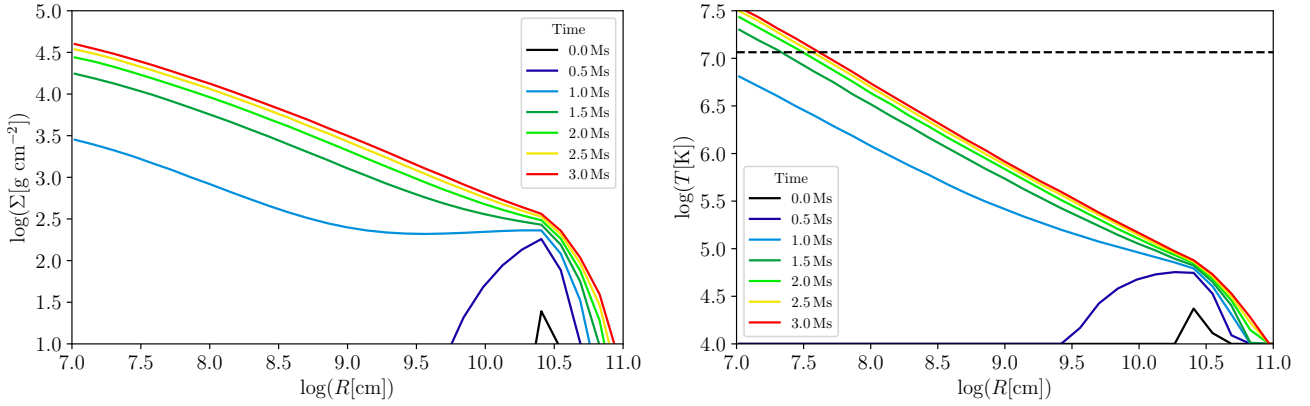
We define a simple fiducial model in which all the matter provided by the Bondi-Hoyle mechanism is effectively accreted ( $\dot{M}_{\text{acc}} = \dot{M}_{\text{acc}}^{\text{BH}}$ ), and the X-ray luminosity of a HMXB is a fraction  $\eta$  of the total accretion power available,

$$l_X = \eta G M_{\text{acc}} \dot{M}_{\text{acc}} R_{\text{acc}}^{-1}, \quad (1)$$

where  $G$  is the gravitational constant. Empirically, the fiducial model overestimates both the size and X-ray luminosity of HMXB populations. To correct this issue, we assume (as usual) that each HMXB emits only during a fraction  $f_{\text{em}}$  of the time, and compute the total X-ray luminosity of a galaxy as

$$L_X = \sum_i f_{\text{em}} l_{X,i}, \quad (2)$$

where we take  $f_{\text{em}}$  as a free model parameter, and the sum is over all the simulated HMXBs in the galaxy.



**Fig. 2.** A typical solution for the unsteady disk flow onto a  $10 M_\odot$  black hole. The surface density (left panel) and temperature (right panel) are shown as a function of radius for different times, increasing from bottom to top. Mass accumulates first near the outer edge, then flows inwards filling the disk until a steady state is reached. The dashed horizontal line corresponds to  $k_B T = 1$  keV, roughly defining the region of the disk that contributes to the X-ray luminosity.

In order to improve the model including the angular momentum transport, we explore the emission of an unsteady accretion disk considering the exchange of mass and momentum with its environment. The scenario is shown in Fig. 1: the compact object accretes through a geometrically thin, optically thick Keplerian disk, which receives matter and momentum at rates  $\dot{M}_{\text{acc}}^{\text{BH}}$  and  $\dot{J}_{\text{acc}}^{\text{BH}}$  respectively. Part of the matter ( $\dot{M}_{\text{acc}}$ ) flows through the disk losing angular momentum and reaching the compact object, whereas the rest is expelled carrying the excess of momentum. The disk spans from the innermost stable circular orbit for black holes or the neutron star radius ( $R_0$ ) to an outer radius  $R_D$ , determined by requiring that  $\dot{J}_{\text{acc}}^{\text{BH}}/\dot{M}_{\text{acc}}^{\text{BH}}$  matches the Keplerian specific angular momentum at  $R_D$ .

We use a finite elements method to integrate the equations governing the flow,

$$\frac{\partial \Sigma}{\partial t} + \frac{1}{R} \frac{\partial}{\partial R} [R \Sigma u] = 0, \quad (3)$$

$$\frac{1}{2} \Sigma u R \Omega + \frac{1}{R} \frac{\partial}{\partial R} [R^2 \Sigma W_{R\phi}] = 0, \quad (4)$$

$$-\frac{3}{2} \Sigma \Omega W_{R\phi} = -\frac{8\sigma}{3\tau} T^4, \quad (5)$$

together with the constitutive relations

$$W_{R\phi} = \alpha m^{-1} k_B T, \quad (6)$$

$$\tau = \frac{\kappa_T \Sigma}{2} + \sqrt{\frac{m}{\pi k_B}} \frac{\beta \Sigma^2 \Omega}{4 T^4}. \quad (7)$$

Here  $t$  is the time,  $R$  and  $\phi$  the radial and azimuth coordinates on the disk, and  $u$ ,  $\Sigma$ ,  $\Omega$ ,  $T$ ,  $W$  and  $\tau$  the radial velocity, surface density, angular velocity, temperature, stress tensor and vertical optical depth of the flow, respectively. These are functions of both  $R$  and  $t$ . The viscosity parameter and mean particle mass of the fluid are respectively,  $\alpha$  and  $m$ ;  $k_B$  and  $\sigma$  are Boltzmann and Stefan-Boltzmann constants. Note that Thomson opacity  $\kappa_T = 0.2(1+X)$  and the radiative opacity parameter  $\beta = 4 \times 10^{25}(1+X)(Z+0.001)$  (both in cgs units) are

explicitly dependent on metallicity  $Z$  and the hydrogen fraction  $X$ .

In Fig. 2 we show the evolution of the surface density and temperature of a typical solution for the disk flow. Both quantities increase first near the outer edge of the disk where matter is injected, and then towards the center as matter flows onto the accretor. The curves reach a steady state consistent with the Shakura & Sunyaev (1973) model in a time scale  $t_{\text{fill}} \sim 10^6$  s, which we compute by fitting an exponential curve to the time evolution of the density near the inner edge of the disk.

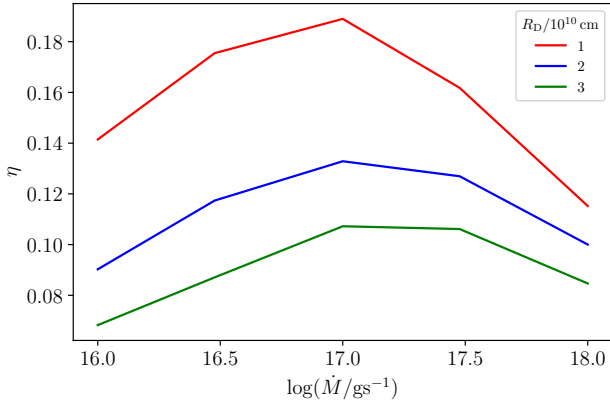
Given the solution for the flow, we compute the X-ray luminosity of the disk in the 0.5 – 8 keV band (that used by Douna et al., 2015) as

$$l_X = 2\pi \int_{R_0}^{R_D} R \int_{0.5 \text{ keV}}^{8 \text{ keV}} B(E; T(R)) dE dR \quad (8)$$

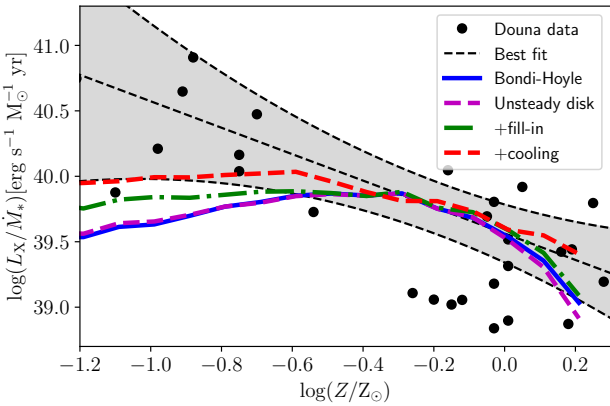
where  $B(E; T)$  is the Planck function depending on the photon energy  $E$  and temperature. Note that our model has no efficiency parameter like  $\eta$ . We couple this model to three different recipes for  $f_{\text{em}}$ : 1) a constant value as in the fiducial model; 2)  $f_{\text{em}} \propto t_{\text{fill}}^{-1}$ ; 3)  $f_{\text{em}} \propto t_{\text{cool}}/t_{\text{fill}}$ , where the cooling time is  $t_{\text{cool}} = k_B T(R_0)/l_X$ . Recipes 2 and 3 assume that the duty cycle of these sources comprises a filling stage, followed by a period in which accretion stops and the disk either is destroyed or cools down. The total galaxy X-ray luminosity is still given by Eq. 2, but in this case  $f_{\text{em}}$  is different for each HMXB.

### 3. Results and discussion

As a first preliminary result, we find that the dependence of the luminosity of unsteady disks on metallicity is negligible. This is due to the fact that the region of the disk with the larger contribution to the X-ray luminosity is at temperatures above  $10^7$  K, at which Thomson opacity dominates. This implies that the observed dependence of the luminosity of HMXB populations on the metallicity of the host galaxy arises from the corresponding dependence of either the parameters of accreting binaries or the size of the population. The latter is



**Fig. 3.** Accretion efficiency  $\eta$  for unsteady disk models as a function of the accretion rate, for different disk radii.



**Fig. 4.** X-ray luminosity of HMXB populations as a function of the host galaxy metallicity. Black circles are the data compiled by Douna et al. (2015); the dashed lines and shaded region mark the best-fitting linear relation and its uncertainty. Model predictions are shown in different colors. We adopted  $Z_{\odot} = 0.0134$  (Asplund et al., 2009).

due to the survival of a larger fraction of systems to the common-envelope phase.

Some authors claim that compact objects have larger masses at low metallicities, resulting in larger luminosities (Mirabel et al., 2011, and references therein). However, we find that the dependence of the model luminosity on the accretor mass is mild ( $\sim 0.2$  dex from 5 to  $100 M_{\odot}$ ). There is instead a strong, almost linear, dependence of the model luminosity with the accretion rate (Fig. 3). The accretion efficiency  $\eta = l_X / (GM_{\text{acc}} \dot{M}_{\text{acc}} R_{\text{acc}}^{-1})$  for our model is almost constant and its value agrees with previous estimations (Frank et al., 2002, and references therein). Note that this value is not a fitted parameter of our model; it arises directly from the physics of our scenario. This trend implies a linear dependence of the luminosity on the accretion rate, the same behaviour observed in the fiducial model.

In Fig. 4 we show the predicted dependence of the X-ray luminosity of HMXB populations  $L_X$  on the metallicity of the host galaxy. Given the aforementioned discussion, the agreement between the fiducial and the un-

steady disk model with a constant  $f_{\text{em}}$  is not surprising. Both agree with the observed trend at solar metallicities, but fail to do so at the low metallicity end. The origin of this disagreement is the strong dependence of the model luminosity on the accretion rate. At low metallicities line-driven winds are less powerful, providing smaller accretion rates. Models that include a physical prescription for  $f_{\text{em}}$  improve the agreement between the unsteady disk predictions and observations. The improvement provided by the filling and cooling times seems to be an effect of the smaller sizes and larger luminosities at lower metallicities.

As a conclusion, we have developed a model for the X-ray luminosity of HMXB populations in galaxies, considering the emission of an unsteady accretion disk. The main preliminary results of this model are the prediction of the value of the accretion efficiency, and the improvement of the description of the observed dependence of the X-ray luminosity of HMXB populations on their host galaxy metallicity when filling and cooling of the disk are taken into account. Future developments include the analysis of the stochasticity of the emission and a more thorough comparison with observations, including population sizes and upper limits to the luminosities in a way similar to that of Douna et al. (2015).

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## References

- Artale M.C., Tissera P.B., Pellizza L.J., 2015, *MNRAS*, 448, 3071
- Asplund M., et al., 2009, *ARA&A*, 47, 481
- Balbus S.A., Hawley J.F., 1998, *Rev.Mod.Phys.*, 70, 1
- Bondi H., Hoyle F., 1944, *MNRAS*, 104, 273
- Bouwens R.J., et al., 2015, *ApJ*, 803, 34
- Bouwens R.J., et al., 2016, *ApJ*, 830, 67
- Brorby M., Kaaret P., Prestwich A., 2014, *MNRAS*, 441, 2346
- Ceverino D., Glover S.C.O., Klessen R.S., 2017, *MNRAS*, 470, 2791
- Ceverino D., Klessen R.S., Glover S.C.O., 2018, *MNRAS*, 480, 4842
- Chabrier G., 2003, *PASP*, 115, 763
- Douna V.M., et al., 2015, *A&A*, 579, A44
- Douna V.M., et al., 2018, *MNRAS*, 474, 3488
- Escobar G.J., Pellizza L.J., Romero G.E., 2021, *A&A*, 650, A136
- Escobar G.J., Pellizza L.J., Romero G.E., 2022, *A&A*, 665, A145
- Fabbiano G., 2006, *ARA&A*, 44, 323
- Feng H., Soria R., 2011, *NewAR*, 55, 166
- Fragos T., et al., 2013, *ApJL*, 776, L31
- Frank J., King A., Raine D.J., 2002, *Accretion Power in Astrophysics: Third Edition*
- Heinz S., Sunyaev R., 2002, *A&A*, 390, 751
- Iorio G., et al., 2023, *MNRAS*, 524, 426
- Kaaret P., Schmitt J., Gorski M., 2011, *ApJ*, 741, 10
- Mirabel I.F., et al., 1992, *Nature*, 358, 215
- Mirabel I.F., et al., 2011, *A&A*, 528, A149
- Ponnada S., Brorby M., Kaaret P., 2020, *MNRAS*, 491, 3606
- Sana H., et al., 2012, *Science*, 337, 444
- Shakura N.I., Sunyaev R.A., 1973, *A&A*, 24, 337